

GETTING STARTED

The AI Lab, and analysis you can defend

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Tue Aug 25, 2026

COURSE OVERVIEW

Two Halves

THE TOOL

Generative AI as it is actually used at work: to write and run code.

You will not write Python.

You will direct it, and check it.

THE SUBJECT

Quantitative investing.
Predicting stock returns, and finding out how little is predictable.

The data is real, and so are the models.

What the Model Is Actually Good For

One claim, and we spend six weeks testing it.

A language model's comparative advantage is turning text into a number. Everything downstream — sizing, portfolio construction, execution — is an optimization problem the model does not help with.

The corollary is the useful part: it is not going to trade for you, but it will write every line of the machinery that does.

Six Weeks

WEEKS	WHAT WE DO
1	Make the machine write code; value a company against its peers
2	Trade on the prediction; regression for multiples
3	Machine learning, and what it does to a valuation model
4	Predicting returns; insider trading and short sales
5	Text as a signal, and trading on sentiment
6	Risk models, and managing portfolio risk

How It Runs

Groups of two or three

Fixed for the term. You work with your group in class, and you present with them.

Everything in the lab

No software on your laptop. A browser and your Rice credentials.

Everything on GitHub

Your group's repository is what survives the term. The lab does not.

Every Class, Three Things

FROM SEPTEMBER 1 ON

- 1 Three groups, drawn at random, present on what they got done last session
- 2 Lecture — the new material for the day
- 3 Hands-on work — you and your group build it

The hands-on portion is the point of the course. The presentations are about it, and the lecture exists to make it possible.

COMPONENT	WEIGHT
Group presentations	60%
Class participation	40%

Groups are drawn at random, so every group should be ready every class. Each group will present twice over the term. Show your repository; there is no need for slides.

What You Are Graded On

By October 1

Your group will have a repository that pulls twenty years of market data, builds a panel of firm characteristics, trains a model on it, and reports honestly whether it works.

Expect the model to explain about one percent of the variance. That is not a bug, and we will spend a full session on why it is a good outcome.

HOW WE GOT HERE

Four Years, Chatbot to Agent

Nov 2022	ChatGPT opens to the public
Mar 2023	GPT-4 and Claude — the chatbot year
Nov 2024	Model Context Protocol — agents get tools
Feb 2025	Claude Code — the agent reaches your files
Oct 2025	Agent Skills — expertise you can package
2026	Cowork and desktop agents

Note where the tools arrive. For the first two years these systems could only talk.

OPENAI CO-FOUNDER, FORMER TESLA AI CHIEF, NOW AT ANTHROPIC

October 2025

“They just don’t work. They don’t have enough intelligence ... You can’t just tell them something and they’ll remember it.”

“The industry is

December 2025

“I really am mostly programming in English now, a bit sheepishly telling the LLM what code to write ... in words.”

“Biggest change to my basic coding

February 2026

“Coding agents basically didn’t work before December.”

“I don’t think I’ve typed like a line of code probably since December.”

Andrej Karpathy

Which Means

The same person, three times in four months.

The version of this you formed an opinion about is probably not the version that exists. Almost everything useful arrived in the last eighteen months, and a good deal of it since last December.

IT WRITES CODE

That Is the Whole Trick

Almost everything useful the model does, it does by writing code and running it.

Usually Python or JavaScript. The code runs on a server — here, on the lab machine — not inside the model.

You are not asking it for an answer. You are asking it for a program that produces an answer, and the program is something you can read.

Excel workbooks with live formulas

Charts and figures

PowerPoint decks

Data analysis and statistics

Word documents

Web apps and dashboards

These are not six features somebody built. They are one capability — write code, run it — pointed in six directions.

One Capability, Six Directions

What That Looks Like Underneath

YOU ASK

Read these four PDFs. Pull out every number either side cites about market share, note which ones disagree, and put it in a table.

WHAT ACTUALLY HAPPENS

Each PDF is converted to text. The text is read. A Python script builds the table. The table is written to a file.

Nothing there is a document feature. It is code execution, pointed at file formats.

Numbers Need Code

THE PROBLEM

To a next-token predictor, “47.3%” is exactly as plausible as “52.1%”.

Mental math, rankings, comparisons, aggregations — all unreliable, and the wrong number arrives with complete confidence.

THE FIX

Insist on code, and insist that it run.

Arithmetic stops being prediction and becomes computation — and you can read afterward what it actually did.

CHATBOTS AND AGENTS

First, a Chatbot

You, a program, and a model. The model reads text and writes text. That is the entire capability.

It cannot open a file, run a calculation, look up today's date, or check whether what it just told you is true.

The chatbot passes along your prompt — plus the system prompt and everything said so far in the conversation — and hands back the reply.

- 1 The agent sends the prompt, the history, and a list of tools it has
- 2 The model answers: use tool X, with these arguments
- 3 The agent runs tool X and reports the result back, with instructions for reading it
- 4 The model decides what to do next — another tool, or an answer for you

The model never runs anything. It emits requests; the agent executes them. Those step-3 instructions are the tool harness, and some are far better than others.

An Agent Is a Chatbot with Tools

THE AI LAB

What You Get

What runs

Claude Code,
running GLM-4.7.

What is installed

Python, JavaScript,
the analysis
libraries, and a set
of Claude Code
plugins.

What you see

A file browser, a
terminal, and a
browser view for
anything you build.

All of it runs in the cloud and is billed to the school.

Sign In

- 1 Go to `lab638.kerryback.com` — any browser, nothing to install
- 2 Username is your netid, the part before `@rice.edu`
- 3 Password is your student number, including the `S0`

First load takes a moment while your machine starts. If you are asked to sign in again later in the term, nothing is wrong — sessions do not survive a restart of the server.

Right panel

Claude Code. The prompt window is at the bottom.

Left panel

File browser by default. Terminal and View are there too — ignore them today.

Three dots, upper left

Upload from your laptop, download to it.

The Workspace

Three Things That Trip Everyone

Refresh

The file browser does not update itself. When GLM makes a file and you cannot see it, hit Refresh.

No Office

Double-clicking a `.xlsx` or `.pptx` does nothing. Office is not installed.

Download to view

Bring Office documents to your laptop and open them there.

WORKING IN GROUPS

Two Places Your Group Lives

A GitHub repository

One shared copy of the code, the notebooks, and the results. It is what you show when you present.

A Slack channel

For coordinating between classes.
Add me to it.

Set both up this week, and send me the repository URL.

Why GitHub

Your group shares one repository, not a folder of attachments.

Mailing notebooks around breaks down by the second week. GitHub keeps one copy that everyone edits, records who changed what and when, and lets two of you work the same afternoon without overwriting each other.

You Do Not Type Git Commands

Same bargain as the code. You say what you want; GLM runs it.

Git has a deserved reputation for cryptic errors. You do not need to learn it — you need to know what you are asking for, and to check that it happened.

There is exactly one step you do yourself, and it is on the next slide.

Once, Per Person

THE ONE THING GLM CANNOT DO FOR YOU

- 1 Create a free account at `github.com`, and note your username
- 2 Open the Terminal tab — not the GLM pane
- 3 Run `gh auth login`, and answer: GitHub.com, HTTPS, log in with a browser
- 4 It prints a code — open the link on your laptop and paste it there

This one waits for you to type, so it has to be the Terminal tab. Asked in the GLM pane, it would sit there until it gave up.

One Person Creates the Repo

TELL GLM

Make my workspace a git repository, commit everything in it, and create a private GitHub repo called mgmt638-team1 from it. Then push.

One person in the group does this, once. Everyone else skips to the slide after next.

Then Add Your Groupmates

ON GITHUB.COM, IN THE NEW REPOSITORY

- 1 Settings, then Collaborators, then Add people
- 2 Search each groupmate by their GitHub username
- 3 They get an email invitation, and have to accept it

Until someone accepts, their clone fails with “not found” rather than “no permission” — the repository is private, so to them it does not exist yet.

Everyone Else Joins

TELL GLM

Clone the GitHub repo `alice/mgmt638-team1` into my workspace and `cd` into it.

`alice` is the GitHub username of whoever created the repo, not yours.

TELL GLM, AT THE START AND END OF EVERY SESSION

Pull the latest from GitHub before we start.

Commit what we did with a sensible message and push it.

Pull first, push last. Nearly every tangle you will hit this term comes from skipping one of the two.

Push and Pull

PROMPTING

Most of the prompt engineering advice from a few years ago is now either useless or actively harmful.

Say what you want, not how to do it

“I want to compare revenue across countries — how should we show it?”

beats

“Make a bar chart, sorted descending, gridlines off, 12pt

If it does not work, try again

Rephrasing is cheap. Treat it as an infinitely patient colleague with a poor memory.

Good Enough Prompting

Plan Before You Act

Use plan mode, or just add: plan how you will do this before you start.

WHY IT WORKS — KNOWLEDGE

Planning pulls the relevant considerations into the conversation. It then answers in light of what it just retrieved, rather than from base rates.

WHY IT WORKS — SCALE

It breaks the work into steps you can check one at a time. Failures stay small and local instead of surfacing at the very end.

Confirm It Understood

Ambiguity does not produce a question. It produces a confident answer to a question you did not ask.

TELL GLM, BEFORE IT STARTS

Tell me what you are about to do and what you are assuming.

Correct it in one sentence, then say go ahead. This is the cheapest habit in the course and it will save you the most time.

It Is a Prediction Machine, Not a Colleague

Suppose it told you last week that method A beats method B. Today you ask it to do the task.

WHAT IT WILL DO

If “A beats B” is not in this conversation, and B appeared more often in its training data, it will probably use B.

WHY

Knowledge in the weights is not knowledge in use. Something has to pull it into the conversation before it can act on it.

Prediction also means momentum. Once it has started down a path, continuing that path is the likeliest next step.

You will watch it hit a roadblock, find a way around, hit another, find another way around — tenaciously, exhaustively. What it rarely does is go back and ask whether a different path was better all along.

Restate the goal, not the obstacle. That gives it permission to abandon the path instead of defending it.

Tunnel Vision

What Not to Do

SKIP	WHY
“Think step by step”	Reasoning models already do, in hidden tokens
“Take a deep breath”	Measurably useless now
“CRITICAL! NEVER EVER ...”	Overtriggers, and gets worse results
Stacked personas	“You are an expert X” can help tone; three of them does not

Direct task framing wins. Say what you want, say what done looks like, say what to avoid.

ANALYSIS, WITHOUT A SPREADSHEET

Go and Get the Data

No file to upload, no API key to paste. Ask.

TELL GLM

Get monthly returns for WMT and SPY since 2020 from Yahoo Finance. Scatter plot, SPY on the horizontal axis, with a regression line.

The slope of that line is Walmart's market beta. You estimated a CAPM regression by asking for a scatter plot.

The code ran on a machine in the cloud. Nobody at Yahoo, and nobody at Anthropic, built a beta calculator.

What You Just Did

Look at the Code

TELL GLM

Show me the code you ran.

You do not need to write Python to read it. Find the line that computes the returns.

Any number that matters comes out of code you can open. That rule applies to every session in this course.

Push It Further

TELL GLM

Add the other Dow components and rank them by beta.

Annotate the three months with the largest residuals and tell me what happened in each.

Watch whether those explanations are sourced or invented.

Put It Somewhere

TELL GLM

Put the chart and a short written summary in a PowerPoint deck.

Then download it and open it on your laptop.

The chart, the layout, the text — all produced by code GLM wrote and ran.

VALIDATION

What Could Go Wrong

You asked for a chart of betas and you got one. Now what?

No answer

Easy. You notice immediately.

Wrong kind of answer

Also easy. A pie chart when you wanted a scatter.

Right kind, wrong numbers

This is the one. It looks exactly like the answer you wanted.

The Trap

Yahoo gives you two closing prices for every day: **Close**, and **Adj Close**, which is adjusted for dividends and splits.

Returns from the wrong one are wrong — and Walmart both pays a dividend and split three-for-one in 2024.

TELL GLM

Recompute the betas both ways and show me the difference.

Misunderstanding

Your instruction was ambiguous.

Ask it to state its plan before it starts.

Silent assumptions

It chose something you never authorized — which price column, which date range.

Ask it to walk through what it did.

Coding errors

Rare. The code either runs or it does not.

Where the Errors Come From

Hallucinations

It is not lying. It is predicting the next token, and a plausible citation is exactly as predictable as a real one.

The risk is highest when the answer is unknown or unfindable — an obscure statistic, a citation that may not exist, a number that was never public. Asked a question with no available answer, it will still produce one.

Make it search

And give you the sources.

Check the sources

Click them. A real-looking citation to a paper that does not exist is the classic

Make it compute

Any number that matters comes out of code.

Training on human feedback teaches that agreement earns high ratings. Ask “is my plan sound?” and it will find reasons that it is.

INVERT THE QUESTION

What is the strongest argument that this recommendation is wrong?

One step further: have it play a specific skeptic — a referee, a regulator, the desk that has to trade this.

Never Ask “Is This Good?”

Set a Critic on It

GLM can dispatch a subagent: a fresh instance, no memory of this conversation.

WHY IT IS WORTH SOMETHING

It sees the instructions, not the reasoning behind the first answer. It has nothing to defend.

Give it a mandate: find every factual error, logical gap, and unsupported

THE CATCH

Two agreeing runs are evidence, not proof. They share a model and they share your ambiguous question.

Three Checks Before You Trust It

Where did this come from?

A citation you can click, a file you can open, or code you can read. If none of the three, treat it as a guess.

What did it assume?

Every unstated choice was made for you. Ask what it decided that you did not specify.

Would I catch it if it were wrong?

If not, that is the part to check by hand — not the part you already understand.

WHO DOES THIS

AQR Capital Management

1998

FOUNDED, OUT OF
CHICAGO AND GOLDMAN

\$189B

UNDER MANAGEMENT

No. 1

LARGEST HEDGE FUND
MANAGER, 2025

A firm whose entire investment process is an argument published in journals.

Background: AQR Capital Management

Quality Minus Junk

Asness, Frazzini and Pedersen, “Quality minus junk”, *Review of Accounting Studies*, 2019.

Watch the video

Quality is one of the characteristics you will put in your own model in week three. This is where it comes from.

BEFORE THURSDAY

Four Things

- 1 Settle your group — two or three people
- 2 Set up the group's GitHub repository and Slack channel, and send me both
- 3 Read the AQR background note
- 4 Bring a question about something GLM got wrong today

Thursday we start on valuation: what a company is worth compared with the companies it looks like.